

an event; instead it is produced via software in real-time. For the interaction to happen, at least a basic recognition system needs to be in place. The most important piece of technology which brings life to the augmentation depends on recognition systems, which we would learn about in a short while.

Location based AR

Next in line in terms of technical complexity is the location based AR. There are multiple factors at play. Location based AR apps take inputs from various sources. The information sources typically include:

- ▶ **GPS** - It allows the app to pinpoint the location of the device.
- ▶ **Acceleration sensor** - to find out how much you moved and in which direction. This prediction allows the app to update the view before the GPS data is received. Also, since accuracy of GPS is limited by approximately a couple of meters, it serves you when you move below that threshold.
- ▶ **Compass** - To tell the direction you are looking in.
- ▶ **Orientation sensor** - It tells the application how much the device is tilted. This helps predict the angle of view from the ground or in other terms, it helps figure out the location and angle of horizon relative to the device camera.

The application helps locate points of interest on your screen by performing the following steps:

- 1 Depending on the location received by the GPS system, the app requests its server to send a list of points-of-interest nearby along with their coordinates.
- 2 It then reads the direction you are looking in (from the compass) and the angle at which you are looking.
- 3 Combining these two pieces of information, it loads the data about the points-of-interest that it received from the server and overlays it on the camera view.

May we remind again that most location based AR apps do not understand what they see – they do not recognize the buildings and mountains (or anything at all). All they do is to calculate your view relative to other points in your surroundings and place them on the screen.

Trigonometry is what puts it all together. The role of AR application is simple – it has to place the information overlay at the correct places. The accuracy of a location based AR app depends on the accuracy of data received from different sensors. If the GPS data is not available, location